The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 October 2005.

INCH-POUND

MIL-PRF-19500/408G
21 July 2005
SUPERSEDING
MIL-PRF-19500/408F
7 November 2003

* PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER, TYPES 2N3715 AND 2N3716 ,JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high-power transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, similar to TO-3.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Types | $P_T$ (1) $T_A$ = +25°C | $P_T$ (2) $T_C$ = +25°C | $R_{\theta JC}$ (3) | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_B$ | $I_C$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | V dc | A dc | A dc | °C |
| 2N3715 | 5.0 | 117 | 1.5 | 80 | 60 | 7.0 | 4.0 | 10 | -65 to +200 |
| 2N3716 | 5.0 | 117 | 1.5 | 100 | 80 | 7.0 | 4.0 | 10 | -65 to +200 |

(1) Derate linearly 28.57 mW/°C above $T_A$ = +25°C.
(2) See figure 2 for temperature-power derating curves.
(3) See figure 3, transient thermal impedance graph.

* 1.4 Primary electrical characteristics. Unless otherwise specified, $T_C$ = +25°C.

| Limit | $h_{FE2}$ (1) $V_{CE}$ = 2.0 V dc $I_C$ = 3.0 A dc | $h_{FE4}$ (1) $V_{CE}$ = 4.0 V dc $I_C$ = 10 A dc | $V_{BE(SAT)1}$ (1) $I_C$ = 5.0 A dc $I_B$ = 0.5 A dc | $V_{CE(SAT)1}$ (1) $I_C$ = 5.0 A dc $I_B$ = 0.5 A dc | $C_{obo}$ $V_{CB}$ = 10 V dc, $I_E$ = 0 f = 1 MHz | $\|h_{fe}\|$ $V_{CE}$ = 10 V dc $I_C$ = 0.5 A dc f = 1 MHz |
|---|---|---|---|---|---|---|
| | | | V dc | V dc | pF | |
| Min | 30 | 5.0 | | | | 4.0 |
| Max | 120 | | 1.5 | 1.0 | 500 | 20 |

(1) Pulsed (see 4.5.1).

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A

FSC 5961

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

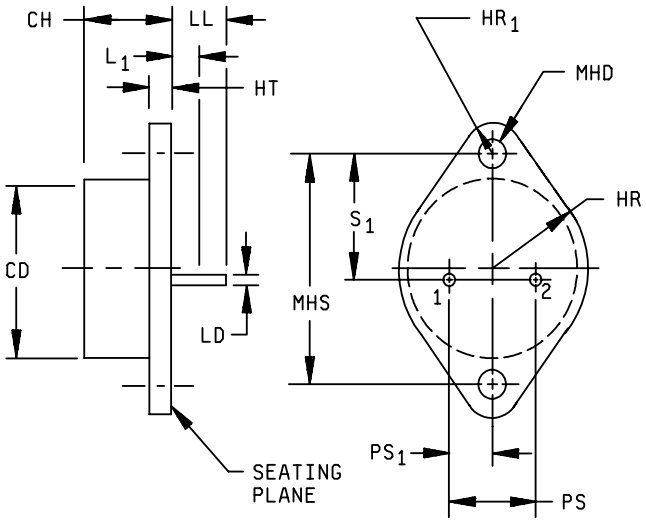
3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.




| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | | .875 | | 22.22 | |
| CH | .270 | .380 | 6.86 | 9.65 | |
| HT | .060 | .055 | 1.52 | 1.40 | |
| HR | .495 | .525 | 12.57 | 13.3 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | |
| LD | .038 | .043 | 0.97 | 1.09 | 7 |
| LL | .312 | .500 | 7.92 | 12.70 | |
| $L_1$ | | .050 | | 1.27 | |
| MHD | .151 | .165 | 3.84 | 4.19 | |
| MHS | 1.177 | 1.197 | 29.90 | 30.40 | |
| PS | .420 | .440 | 10.67 | 11.18 | 4 |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | 4 |
| $s_1$ | .655 | .675 | 16.64 | 17.15 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1, emitter; terminal 2, base; case, collector.
4. These dimensions should be measured at points .050-.055 inch (1.27-1.40 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .004 inch (0.10 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .006 inch (0.15 mm) concave to .006 inch (0.15 mm) convex overall.
6. Collector shall be electrically connected to the case.
* 7. LD applies between $L_1$ and LL. Lead diameter shall not exceed twice LD within $L_1$.
8. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

* FIGURE 1. Physical dimensions of transistor types 2N3715 and 2N3716 (TO-3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance (transient), method 3131 of MIL-STD-750 (see 4.3.2). | Thermal impedance (transient), method 3131 of MIL-STD-750 (see 4.3.2). |
| 7 | Optional. | Optional. |
| 9 | $I_{CES1}$ and $h_{FE2}$ | $I_{CES1}$ |
| 11 | $I_{CES1}$ and $h_{FE2}$;<br>$\Delta I_{CES1}$ = 100 percent of initial value or 1 μA dc, whichever is greater.<br>$\Delta h_{FE2}$ = ± 15 percent of initial value. | $I_{CES1}$ and $h_{FE2}$;<br>$\Delta I_{CES1}$ = 100 percent of initial value or 1 μA dc, whichever is greater. |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | $\Delta I_{CES1}$ = 100 percent of initial value or 1 μA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ± 15 percent of initial value;<br>subgroup 2 of table I herein | $\Delta I_{CES1}$ = 100 percent of initial value or 1 μA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ± 20 percent of initial value;<br>subgroup 2 of table I herein |
| 14 | Required. | Required. |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_J$ = 187.5 ± 12.5°C; $V_{CE}$ = 25 ± 5 V dc; $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). The thermal impedance limit used in screen 3c of 4.3 herein and table I shall comply with the thermal impedance graph in figure 3 (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection . Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 30 V dc, $P_T$ = 4 W at $T_A$ = +25°C ± 3°C, $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted. A separate sample may be pulled for each test. |
| B5 | 1027 | $V_{CB}$ = 30 V dc, $T_A$ = +125°C ± 25°C for 96 hours; $P_T$ = 4 W at $T_A$ = +100°C or adjusted as required by the chosen $T_A$ to give an average lot $T_J$ = +225°C. Marking legibility requirement shall not apply. |

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1037 | $V_{CB} \geq$ 10 V dc; $\Delta T_J$ between cycles ≥ 100°C; $t_{on}$ = $t_{off}$ ≥ 1 minute for 2,000 cycles. No heat sink or forced air cooling on the device shall be permitted. |
| B6 | 1032 | $T_A$ = +200°C. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 1056 | Test condition B. |
| C2 | 2036 | Test condition A, weight = 10 lbs, application time = 15 seconds. |
| * C5 | 3131 | See 4.5.2. |
| C6 | 1037 | $V_{CB} \geq$ 10 V dc; $\Delta T_J$ between cycles ≥ 100°C, $t_{on}$ = $t_{off} \geq$ 1 minute for 6,000 cycles. No heat sink or forced air cooling on the device shall be permitted. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply.

a. Collector current magnitude during power application shall be 1.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be within the range $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC} = 1.5°C/W$ and shall be in accordance with figure 3, thermal impedance graph.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| * Thermal impedance 2/ | 3131 | See 4.3.2 | $Z_{\theta JC}$ | | | °C/W |
| Collector-emitter breakdown voltage | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N3715 | | | | 60 | | V dc |
| 2N3716 | | | | 80 | | V dc |
| Collector-base cutoff current | 3036 | Base condition D | $I_{CBO}$ | | | |
| 2N3715 | | $V_{CB}$ = 80 V dc | | | 10 | µA dc |
| 2N3716 | | $V_{CB}$ = 100 V dc | | | 10 | µA dc |
| Emitter-base cutoff current | 3061 | Base condition D; $V_{EB}$ = 7 V dc | $I_{EBO}$ | | 1.0 | mA dc |
| Collector-emitter cutoff current | 3041 | Bias condition A | $I_{CEX1}$ | | | |
| 2N3715 | | $V_{BE}$ = 1.5 V dc, $V_{CE}$ = 60 V dc | | | 1.0 | mA dc |
| 2N3716 | | $V_{BE}$ = 1.5 V dc, $V_{CE}$ = 80 V dc | | | 1.0 | mA dc |
| Collector-emitter cutoff current | 3041 | Bias condition C | $I_{CES1}$ | | | |
| 2N3715 | | $V_{CE}$ = 60 V dc | | | 1.0 | mA dc |
| 2N3716 | | $V_{CE}$ = 80 V dc | | | 1.0 | mA dc |
| Base-emitter saturated voltage | 3066 | Test condition A; $I_C$ = 5 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{BE(SAT)1}$ | | 1.5 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A; $I_C$ = 10 A dc; $I_B$ = 2 A dc; pulsed (see 4.5.1) | $V_{BE(SAT)2}$ | | 3.0 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 5 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{CE(SAT)1}$ | | 1.0 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 10 A dc; $I_B$ = 2 A dc; pulsed (see 4.5.1) | $V_{CE(SAT)2}$ | | 2.5 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 1.0 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 50 | 150 | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 3.0 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 30 | 120 | |

See footnote at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

| | Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| | Subgroup 2 - Continued | | | | | | |
| | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 5 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 10 | | |
| | Forward-current transfer ratio | 3076 | $V_{CE}$ = 4.0 V dc; $I_C$ = 10 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | 5 | | |
| | Subgroup 3 | | | | | | |
| | High temperature operation: | | $T_A$ = +150°C | | | | |
| | Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES2}$ | | | |
| | 2N3715 | | $V_{CE}$ = 50 V dc | | | 5.0 | mA dc |
| | 2N3716 | | $V_{CE}$ = 70 V dc | | | 5.0 | mA dc |
| | Low temperature operation: | | $T_A$ = -55°C | | | | |
| * | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 3.0 A dc; pulsed (see 4.5.1) | $h_{FE5}$ | 12 | | |
| | Subgroup 4 | | | | | | |
| * | Pulse response: | 3251 | Test condition A, except test circuit and pulse requirement in accordance with figure 4 | | | | |
| | Pulse delay time | | | $t_d$ | | 0.2 | µs |
| | Pulse rise time | | | $t_r$ | | 1.3 | µs |
| | Pulse storage time | | | $t_s$ | | 1.2 | µs |
| | Pulse fall time | | | $t_f$ | | 1.2 | µs |
| | Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc; f = 1 kHz | $h_{fe}$ | 30 | 300 | |
| | Magnitude of small-signal short-circuit forward current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc; f = 1 MHz | $\|h_{fe}\|$ | 4.0 | 20 | |
| | Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; f = 1 MHz | $C_{obo}$ | | 500 | pF |

See footnote at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc)<br>Test 1 | 3051 | $T_C$ = +25°C; t ≥ 1 s; 1 cycle; (see figure 5)<br>$V_{CE}$ = 15 V dc; $I_C$ = 10 A dc | | | | |
| Test 2 | | $V_{CE}$ = 40 V dc; $I_C$ = 3.75 A dc | | | | |
| Test 3<br>2N3715<br>2N3716 | | <br>$V_{CE}$ = 55 V dc; $I_C$ = 0.9 A dc<br>$V_{CE}$ = 65 V dc; $I_C$ = 0.9 A dc | | | | |
| Safe operating area (clamped inductive) | 3053 | $T_A$ = +25°C; $I_C$ = 10 A dc;<br>$V_{CC}$ = 15 V dc; (see figures 6 and 7) | | | | |
| 2N3715<br>2N3716 | | Clamp voltage = 60 V dc<br>Clamp voltage = 80 V dc | | | | |
| Test 1 | | $t_p$ approximately 1.0 ms (vary to obtain $I_C$); $R_{BB1}$ = 5Ω; $V_{BB1}$ = 8 V dc; $R_{BB2}$ = 4; $V_{BB2}$ = 0; $V_{CC}$ = 15 V dc; $I_C$ = 5 A dc; L = 4 mH; Q ≥ 100 at 1 kHz; .05 ohms, 20 A dc; (Stanford Miller CK-20 or equivalent) | | | | |
| Test 2 | | $t_p$ approximately 2.2 ms (vary to obtain $I_C$); $R_{BB1}$ = 5Ω; $V_{BB1}$ = 8 V dc; $R_{BB2}$ = 4; $V_{BB2}$ = 0; $V_{CC}$ = 15 V dc; $I_C$ = 2.2 A dc; L = 20 mH; Q ≥ 6000 at 10 kHz; 0.22 ohms, 12.5 A dc; (Two Stancor C-2688 in series or equivalent) | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector-emitter cutoff current<br>2N3715<br>2N3716 | 3041 | Bias condition C<br><br>$V_{CE}$ = 60 V dc<br>$V_{CE}$ = 80 V dc | $\Delta I_{CES1}$ 5/ | 100 percent of initial value or 1 μA dc, whichever is greater. | | |
| 2 | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc;<br>$I_C$ = 3.0 A dc;<br>pulsed (see 4.5.1) | $\Delta h_{FE2}$ 5/ | ± 25 percent change from initial value | | |
| 3 | Collector-emitter (voltage saturated) | 3071 | $I_C$ = 5 A dc;<br>$I_B$ = 0.5 A dc;<br>pulsed (see 4.5.1) | $\Delta V_{CE(SAT)1}$<br><br>5/ | ± 50 percent mV dc change from previously measured value. | | |

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table II herein, step 2.
   b. Subgroup 5, see table II, steps 1, 2, and 3.

2/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows: Subgroups 3 and 6, see table II herein, steps 1 and 2.

3/ The delta measurements for table VII of MIL-PRF-19500 are subgroup 6, see table II herein, all steps for JANS and steps 1 and 2 for JAN, JANTX, and JANTXV.

4/ The delta measurements for table IX of MIL-PRF-19500 are as follows: Subgroups 1 and 2, see table II herein, all steps.

5/ Devices which exceed the group A limits for this test shall not be acceptable.



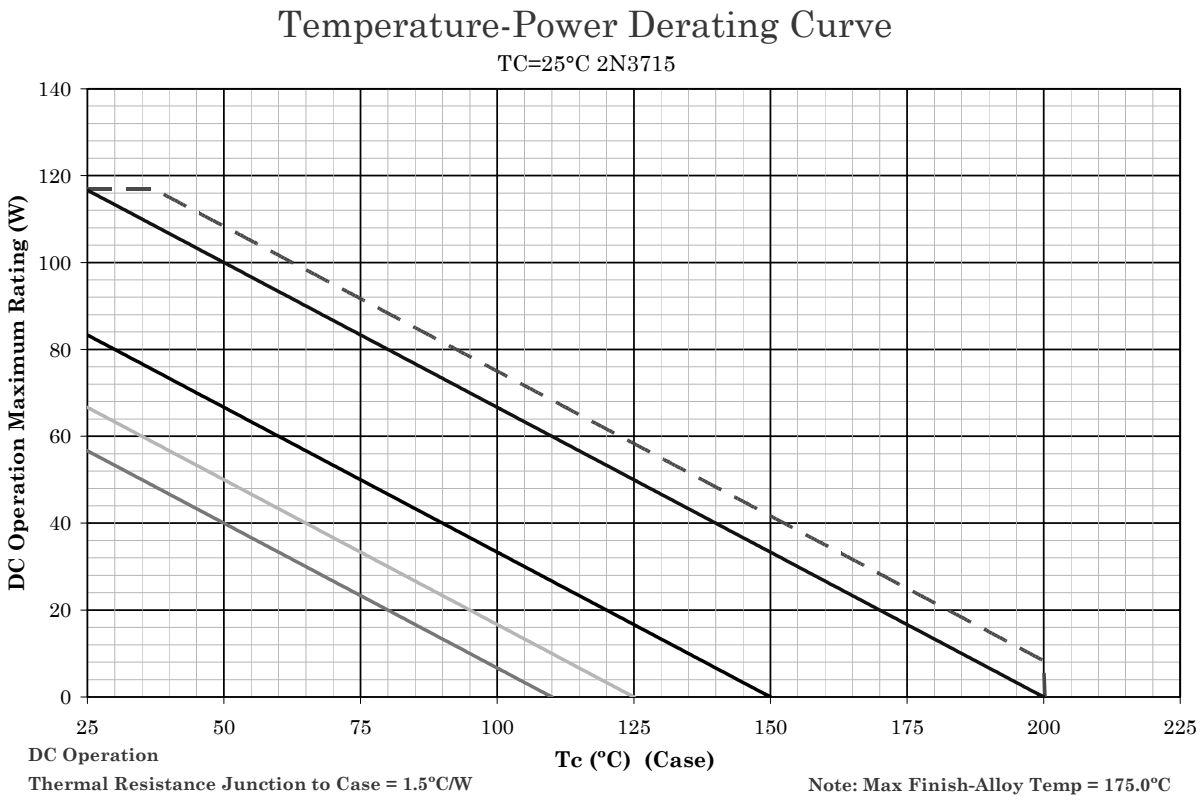
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

* TABLE III. Group E inspection (all quality levels) - for qualification and re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling | 1051 | 500 cycles minimum | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and table II. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Blocking life | 1048 | Test temperature = +125°C; $V_{CB}$ = 80 percent of rated voltage, T = 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2 and table II. | |
| Subgroup 4 | | | sample size<br>N/A |
| Thermal impedance curves | | Each supplier shall submit their qual-lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| ESD | 1020 | Testing not required for class 3 listing. Testing is required for nonsensitive listing to prove capability. | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.



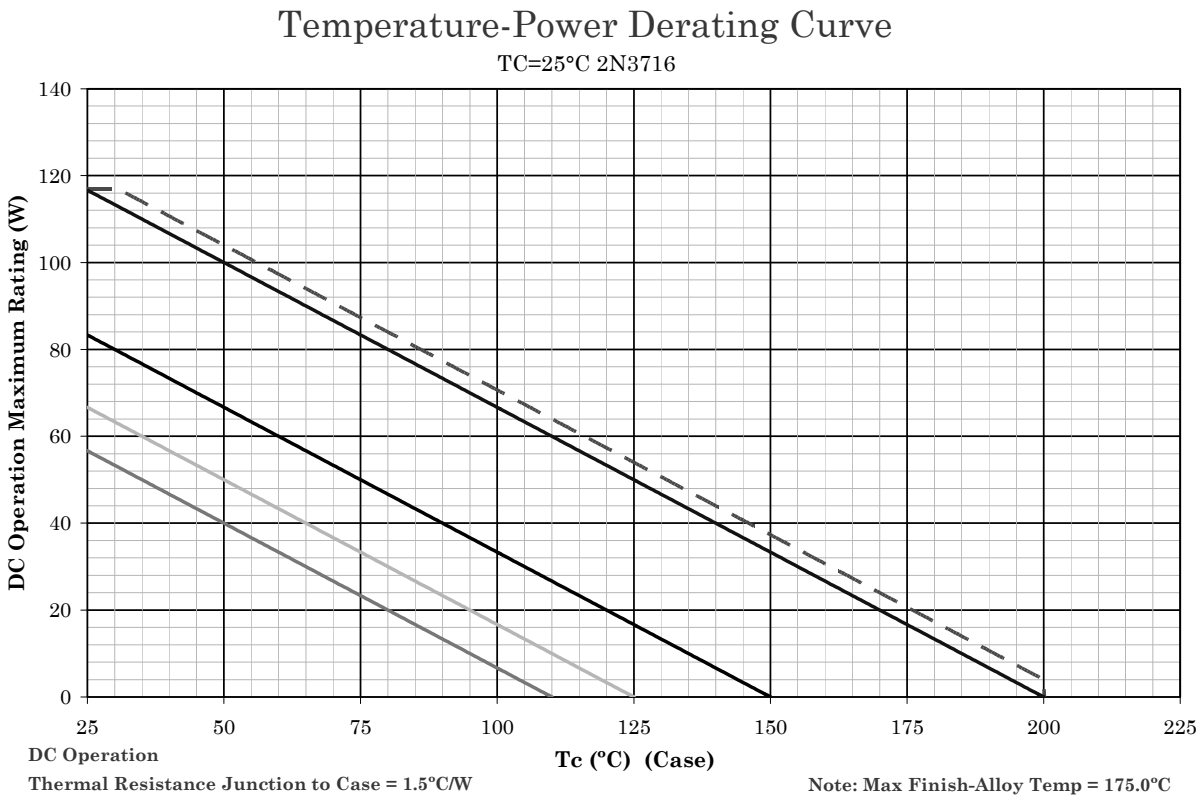

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves, chosen at $T_J \leq +125°C$ and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 2. Temperature derating graphs (TO-3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.



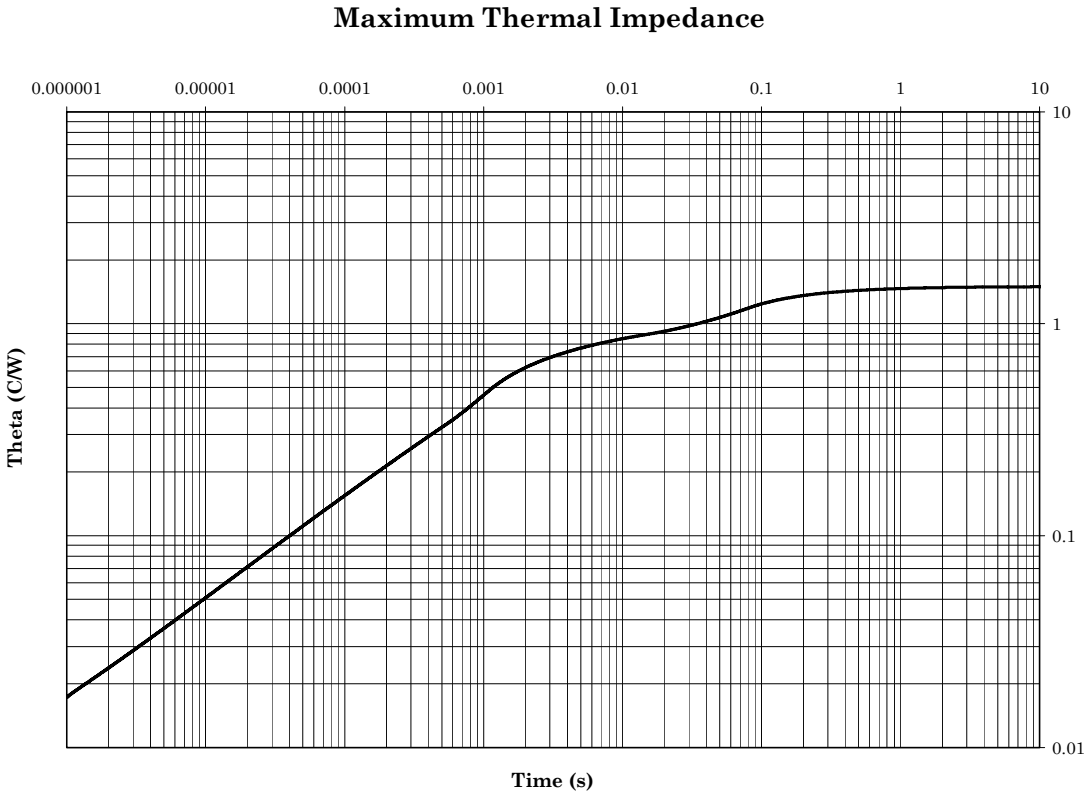

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves, chosen at $T_J \leq +125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 2. Temperature derating graphs (TO-3) - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

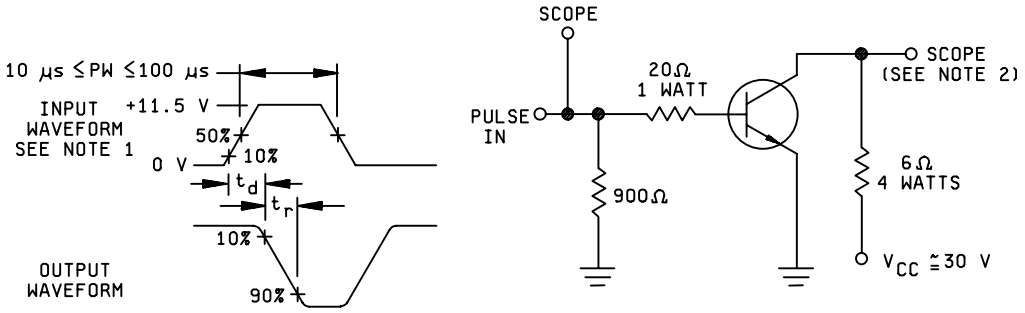



$T_C$ = +25C. Thermal resistance = 1.5°C/W.

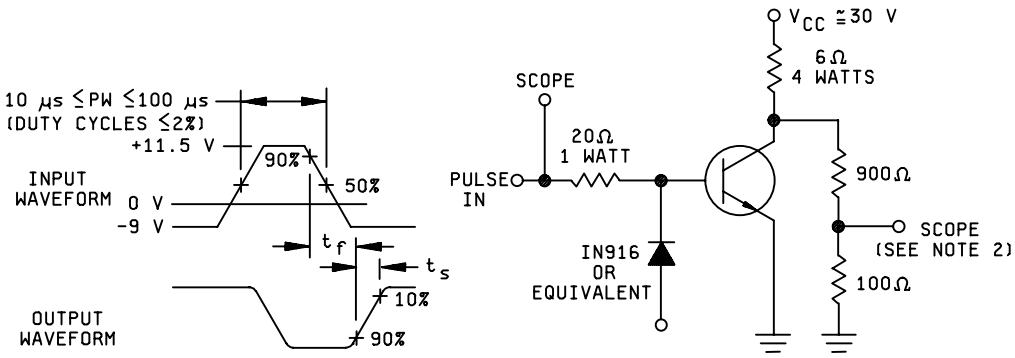
FIGURE 3. Transient thermal impedance graph.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.




TEST CIRCUIT A




TEST CIRCUIT B

NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20.0$ ns, $t_f \leq 1$ µs, 10 µs ≤ PW ≤ 100 µs, $Z_{OUT} = 50\Omega$, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 5$ ns, $Z_{IN} \geq 100$ kΩ, $C_{IN} \leq 12$ pF.
3. Test circuit A for $t_d$ and $t_r$; test circuit B for $t_s$ and $t_f$.

FIGURE 4. Pulse response test circuits.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

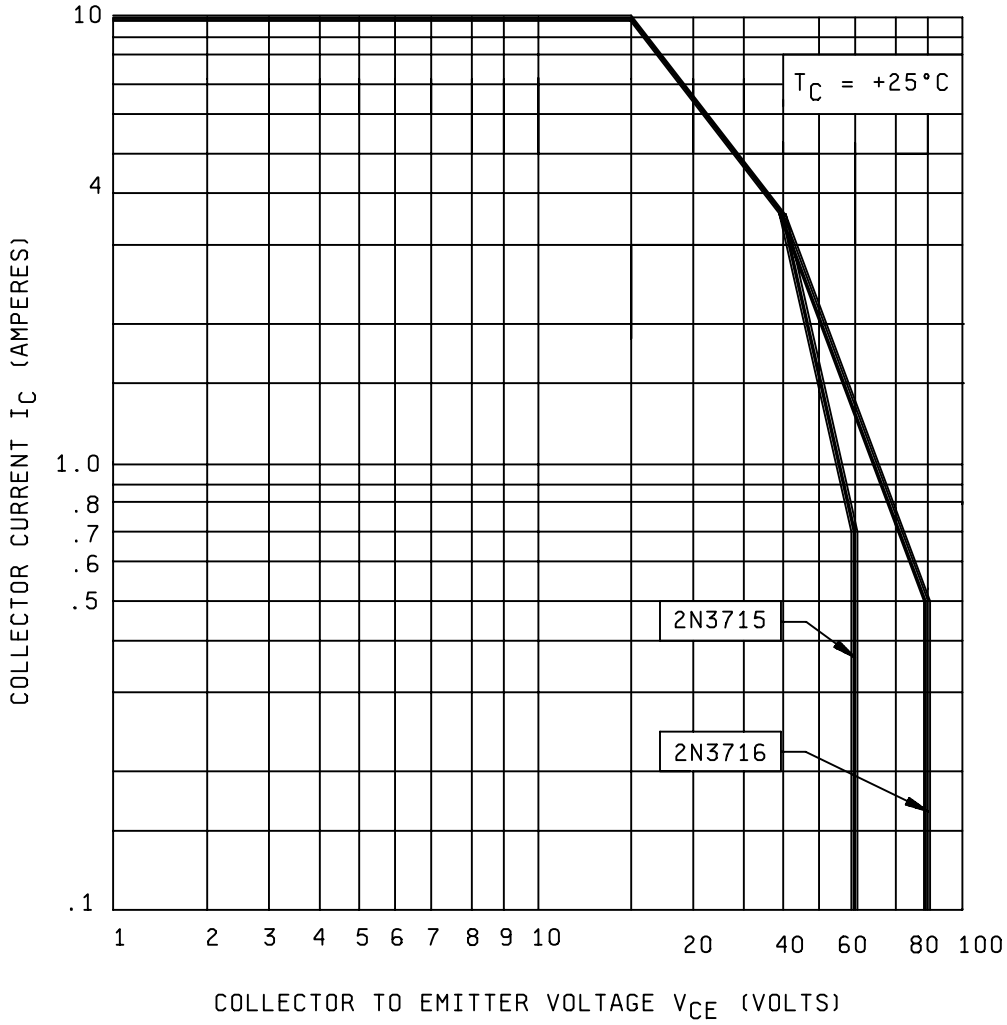



FIGURE 5. Maximum safe operating graph (dc).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

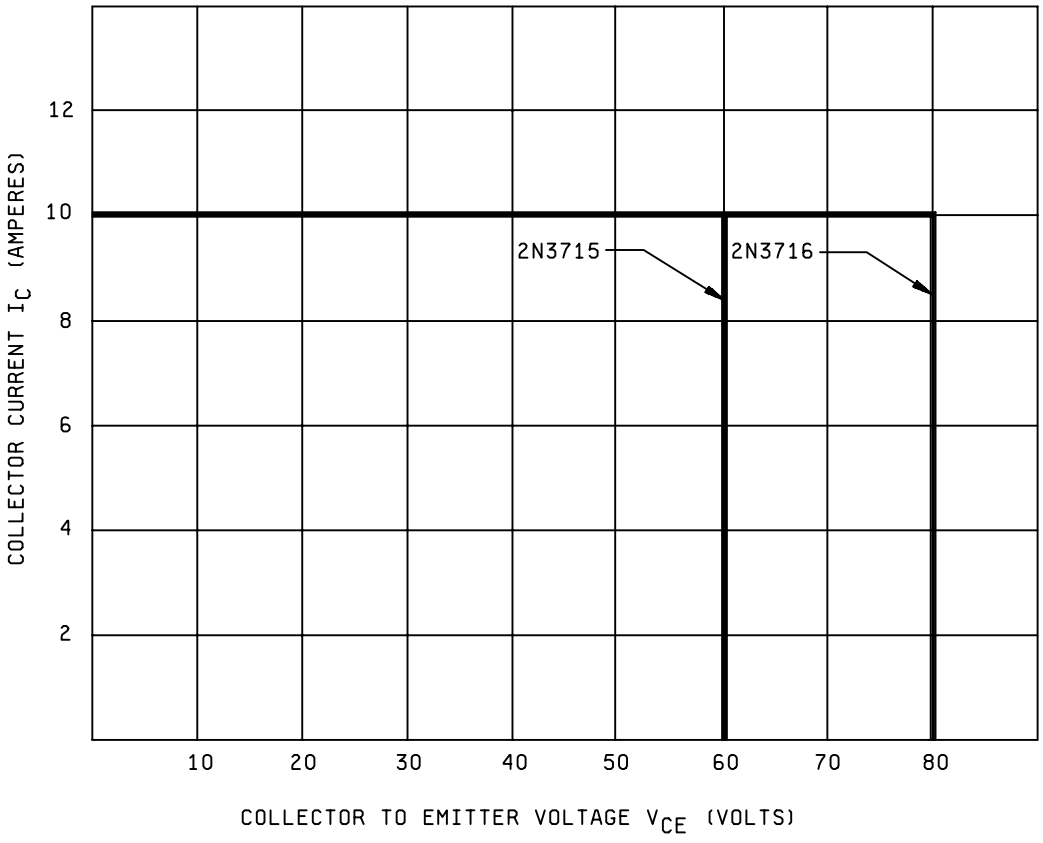
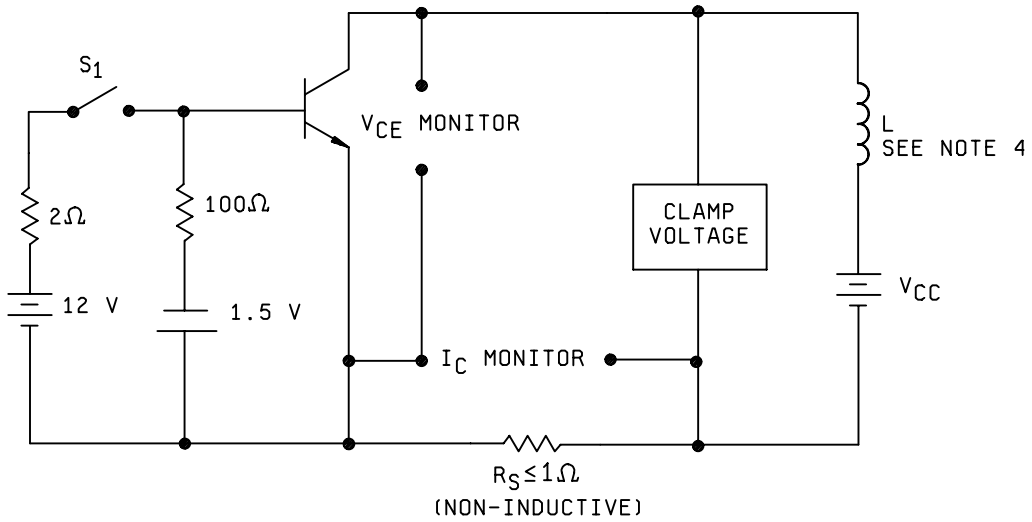



FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.




Procedure:

1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached.
3. Perform specified end-point tests.
4. L = 4.0 mH, 0.05Ω, 20 A, Q ≥ 100 at 1 kHz, (Sanford Miller CK-20 or equivalent).

FIGURE 7. Clamp inductive sweep test circuit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

* 6.4 Interchangeability information. MIL-PRF-19500/622 is a TO-254 package version of MIL-PRF-19500/408, which is a TO-3 package version. The military 2N7368 contains the same die as the military 2N3716. The MIL-PRF-19500/622 is preferred over the MIL-PRF-19500/408 whenever interchangeability is not a problem. For new design use 2N7368 instead of 2N3716. The 2N3716 only is inactive for new design.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2950)

Review activities:
Army - AR, AV, MI
Navy - AS, MC
Air Force - 19, 99

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:39Z
Check the source to verify that this is the current version before use.